

This is the first trial of my Gemini Gem. I uploaded a pdf of math4ml that I had frequently referenced in the beginning. It is linked here: <https://gwthomas.github.io/docs/math4ml.pdf>



Help! I dont know where to start with induced rms-rms norm.

P Prerequisite Preparator • Custom Gem

Show thinking ▼

It sounds like you're diving into the mechanics of normalization layers! Let's clear up the **Induced RMS-RMS norm** first, and then we'll jump straight into the rapid-fire study session.

It began with a good introduction of induced rms-rms norm, upon my request.

Induced RMS-RMS Norm

The "induced RMS-RMS norm" is a matrix norm derived by equipping both the input and output vector spaces with the **Root Mean Square (RMS) norm** instead of the standard Euclidean (L_2) norm.

1. **The Vector RMS Norm:** For a vector $x \in \mathbb{R}^d$, the RMS norm is the quadratic mean of its entries, which relates to the Euclidean norm ($\|x\|_2$) by a factor of the square root of the dimension d :

$$\|x\|_{RMS} = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2} = \frac{1}{\sqrt{d}} \|x\|_2$$

 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)

$\|x\|_2$ by a factor of the square root of the dimension d .



$$\|x\|_{RMS} = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2} = \frac{1}{\sqrt{d}} \|x\|_2$$

2. **The Induced Matrix Norm:** An induced matrix norm measures the maximum factor by which a matrix W (shape $m \times n$) can "stretch" a vector, relative to the norms chosen for the domain (n) and codomain (m).

$$\|W\|_{RMS \rightarrow RMS} = \sup_{x \neq 0} \frac{\|Wx\|_{RMS}}{\|x\|_{RMS}}$$

3. **The Result:** Substituting the vector definitions, we get:

$$\|W\|_{RMS \rightarrow RMS} = \sup_{x \neq 0} \frac{\frac{1}{\sqrt{m}} \|Wx\|_2}{\frac{1}{\sqrt{n}} \|x\|_2} = \sqrt{\frac{n}{m}} \left(\sup_{x \neq 0} \frac{\|Wx\|_2}{\|x\|_2} \right)$$

Since the term in the parenthesis is the **Spectral Norm** ($\|W\|_2$, or the largest singular value σ_{max}), the induced RMS-RMS norm is simply the spectral norm scaled by the ratio of the square roots of the input and output dimensions (often called fan-in and fan-out).

 Enter a prompt for Gemini

  Tools

Thinking 



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)

$$\|v\|_{RMS \rightarrow RMS} = \sqrt{\text{fan-out} \cdot \sigma_{\max}(v)}$$



Rapid-Fire Matrix Algebra

Let's begin. I will ask one question. Answer briefly, and I'll verify it and give you the next one.

It finishes its explanation with a quiz-style question, to be asked one at a time. This is like how I specified in the Gem setup.

Question 1: Since we just established that the spectral norm is the largest singular value ($\sigma_{\max}$), what is the relationship between the **singular values** of a real matrix A and the **eigenvalues** of the symmetric matrix $A^T A$?

the singular values are the square root of the eigenvalues of the symmetric matrix?

P Prerequisite Preparator • Custom Gem

Show thinking ▼

Correct!

 Enter a prompt for Gemini



Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)

 Gemini




Prerequisite Preparator • Custom Gem



Show thinking ▼

Correct!

To be precise, the singular values σ_i of matrix A are the square roots of the eigenvalues λ_i of the Gram matrix $A^T A$. 

$$\sigma_i(A) = \sqrt{\lambda_i(A^T A)}$$

Question 2: If a symmetric matrix A is **Positive Semi-Definite (PSD)**, what condition must **all** of its eigenvalues satisfy?

 Sources

 they must be nonnegative

P Prerequisite Preparator • Custom Gem

Show thinking ▼

 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)



P Prerequisite Preparator • Custom Gem

Show thinking ▼

Spot on. $\lambda_i \geq 0$.

Question 3: In terms of the quadratic form, for a real symmetric matrix A to be **Positive Semi-Definite**, what inequality must the scalar $x^T Ax$ satisfy for **any** vector $x \in \mathbb{R}^n$?

≥ 0

P Prerequisite Preparator • Custom Gem

Show thinking ▼

Correct. $x^T Ax \geq 0$.

Question 4: If a square matrix Q is **orthogonal**, what is its inverse Q^{-1} equal to?

At this point, I thought that the questions it was asking me were way too introductory. This wasn't helping me with my learning.

 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)



Correct. $x^T Ax \geq 0$.

Question 4: If a square matrix Q is **orthogonal**, what is its inverse Q^{-1} equal to?

My apologies, there is some overlapping response between this page and the last; I'm having pdf troubles.

As i mentioned on the previous page, I thought these questions were too elementary to be truly useful. I thought maybe if I switch the topic it would ask different, more difficult questions.

Q^T .

can you now ask me questions about another important topic?

P Prerequisite Preparator • Custom Gem

Show thinking ▼

Correct! For an orthogonal matrix, the transpose is the inverse: $Q^{-1} = Q^T$.

Question 5: For any square matrix A , the **trace** ($\text{tr}(A)$) is the sum of its diagonal elements. What is the trace equal to in terms of the matrix's **eigenvalues**?

Still no, this isn't the kind of preparation I was envisioning. After this, I made a change to the gem: I uploaded the solutions and questions to homework 0, so it could directly reference concepts tested in prerequisite homework for CS182 specifically. The next page begins my prompts, with the new Gem having updated information.

 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)



I'm not sure where to start in making sure I have the necessary prerequisite knowledge, can you help me?

P Prerequisite Preparator • Custom Gem

Show thinking ▼

I can certainly help you with that. The "Mathematics for Machine Learning" document you uploaded is an excellent guide for this. I'll use it to quiz you on the core concepts required for your Deep Neural Networks class.

Let's jump right in with a fundamental concept often used in dimensionality reduction and optimization.

Question 1: According to the **Spectral Theorem**, what specific properties do the **eigenvalues** and **eigenvectors** of a real, symmetric matrix always possess?

Success; i think the questions are now a bit more interesting. I didn't really like how it mentioned the document I uploaded, it's supposed to be a stand alone study tool. But I understand this is how it works and upon further consideration, I'd honestly prefer for it to be transparent.

🛡 Enter a prompt for Gemini



🛠 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)



can you ask questions more tailored to the "hw00" solutions I uploaded?

P Prerequisite Preparator • Custom Gem

Show thinking ▼

I've analyzed the `hw00_solution.pdf` you uploaded. It covers vector calculus identities, SVD interpretations of least squares (both over- and under-determined), and various perspectives on Ridge Regression.

Let's start with the **Vector Calculus** basics from Section 2.

Question 1: Deriving the gradient of a quadratic form is a staple operation. In general, what is the derivative with respect to vector x of the scalar quantity $x^T Ax$?

(Note: Your homework specifies a convention where derivatives of scalars are row vectors.)

🛡 Enter a prompt for Gemini

+ 🛠 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)

 Gemini


The rest of this conversation, I continue to ask it questions. I also test to see if it can identify nuances; e.g. here, $2x^t a$ is only true if A is a symmetric matrix. Indeed it caught the answer. I appreciate its wording ("Close!") it is an encouraging and educational tone that is helpful.

 $2x^t a$

P Prerequisite Preparator • Custom Gem

Show thinking ▼

Close! That is the correct answer **if and only if** the matrix A is **symmetric** ($A = A^T$). 

In the general case (where A may not be symmetric), the derivative is $x^T(A + A^T)$.  

Question 2: Moving on to Least Squares:

If you have an **overdetermined** system $Xw = y$ (where X is a tall matrix with more rows than columns), what is the closed-form solution for the weights $\hat{w}$ that minimize the squared error $\|Xw - y\|^2$?

 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)



We can use the min-norm solution

P Prerequisite Preparator • Custom Gem

Show thinking ▼

Actually, that's not quite right for the **overdetermined** case.

The **min-norm solution** is used when the system is **underdetermined** (wide matrix, $m < n$), where there are infinite solutions and we want the one with the smallest norm. [↗](#)

For the **overdetermined** (tall matrix) case I asked about, we look for the **Least Squares** solution. [↗](#)

The correct closed-form solution is:

$$\hat{w} = (X^T X)^{-1} X^T y$$



 Enter a prompt for Gemini



 Tools

Thinking ▼



Your UC Berkeley chats aren't used to improve our models. Gemini can make mistakes, so double-check it.
[Your privacy & Gemini](#)